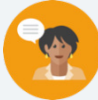


10.H2 Absolute Value Inequalities in Real-World Contexts

Lesson Introduction

 Benchmark	MA.912.AR.4.2 Given a mathematical or real-world context, write and solve one-variable absolute value inequalities. Represent solutions algebraically or graphically.		 Learning Targets	- I can write and solve absolute value inequalities from a real-world context. - I can represent solutions to absolute value inequalities algebraically or graphically.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can I solve absolute value inequalities from a real-world context? - How can I represent solutions to absolute value inequalities algebraically? - How can I represent solutions to absolute value inequalities graphically?			
 Lesson Narrative	This lesson is the second of a two-part series on absolute value inequalities. Students begin by collaborating on real-world absolute value inequalities. Students will then be formally introduced to real-world absolute value inequalities. They will continue practicing this concept both collaboratively and independently.			
 Component Summary	10.H2.1 Warm-Up (~5 min.)	Absolute value equation Error Analysis activity (<i>SE p. 148</i>)	10.H2.4 Guided Practice (10 min.)	Practice with real-world absolute value inequalities (<i>SE p. 150</i>)
	10.H2.2 Collaboration (10-15 min.)	Collaborative activity with real-world absolute value inequalities (<i>SE p. 148</i>)	10.H2.5 Your Turn! (5 min.)	Independent practice with real-world absolute value inequalities (<i>SE p. 151</i>)
	10.H2.3 Guided Instruction (10 min.)	Formal introduction with real-world absolute value inequalities (<i>SE p. 149</i>)	10.H2.6 Wrap-Up (5 min.)	Learning target reflection activity (<i>SE p. 152</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Absolute Value <u>Additional Terms:</u> N/A		N/A	
			Additional Preparation	
			N/A	

	Common Misconceptions	Things to Consider
 Teacher Tips	Students may believe “larger” negative numbers are greater in value, therefore incorrectly graphing inequalities.	- This is an honors lesson that is exclusively for students enrolled in Algebra 1 Honors. Consider moving this lesson to post-assessment as a capstone experience if curriculum pacing is a concern. - If time is an issue, then consider assigning Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson provides students the opportunity to collaborate on real-world applications of absolute value functions. For this lesson, consider leveraging the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.	MA.K12.MTR.4.1: Discussion and Reflection This lesson consists solely of real-world problems. During this lesson, students will connect mathematical concepts to everyday experiences. These problems will provide opportunities to create models, both concrete and abstract, and perform investigations.
 Differentiate Instruction	10.H2.4 Guided Practice incorporates a variety of concepts from throughout the unit and acts as a culminating demonstration of students’ learning progression. Consider what differentiating could look like to maximize student application. - Instead of leading a small group, consider having a student lead a small group. - Prompt students to create their own inequality, before trading with a partner to solve. - Encourage students to represent each question using a visual, using models or diagrams, to represent the situation.	
Lesson Notes:		

10.H2.1 Warm-Up: Error Analysis

Component Overview: Students will complete an Error Analysis activity with an absolute value equation.



10.H2.1 Warm-Up: Error Analysis

Ethan's work for solving $10 - 5|x - 4| = 45$ is shown.

$$\begin{array}{l}
 \textcircled{10} - 5|x - 4| = 45 \\
 5|x - 4| = 45 \\
 \frac{5|x - 4|}{5} = \frac{45}{5} \\
 |x - 4| = 9 \\
 \begin{array}{r}
 x - 4 = 9 \\
 +4 \quad +4 \\
 \hline
 x = 13
 \end{array}
 \quad
 \begin{array}{r}
 x - 4 = -9 \\
 +4 \quad +4 \\
 \hline
 x = -5
 \end{array}
 \end{array}$$



This activity highlights the trademarks of MA.K12.MTR.4.1. During this activity, students will analyze the mathematical thinking of others, recognize errors, suggest how to correctly solve the task, and justify results by explaining methods and processes.

1. Circle the mistake Ethan made.

Answer above

2. Find the correct solution to $10 - 5|x - 4| = 45$.

No solution

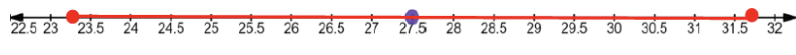
10.H2.2 Collaboration: Real-World Absolute Value Inequalities

Component Overview: Students will work with a partner to collaborate on a multipart real-world problem involving absolute value inequalities.



10.H2.2 Collaboration: Real-World Absolute Value Inequalities

1. Members of the cycling team are preparing for a ten-mile time trial. The average time for the team is 27.5 minutes. The fastest and slowest times varied from the average by 4.2 minutes.
 - a. Based on the given information, what is a time for the team that is not likely?
Answers will vary, but should be greater than 31.7 or less than 23.3 minutes.
 - b. Discuss with your partner if it is possible that a team had a trial time of 30 minutes. Summarize your discussion.
It is possible for the team to achieve an average time of 30 minutes based on the possible team times.
 - c. Work with your partner to show on the number line all the possible values for the time it can take the team to complete the time trial.



The shaded region represents all times that are the possible distances away from the average time.



Absolute value is a number's distance from zero on a number line. Distance is expressed as a positive value.

In this case, the distance is not from 0 but from 27.5. If x is the time for a ten-mile trial, then $|x - 27.5|$ represents the distance that the time, x , is from 27.5.



Three Reads

For this lesson, consider leveraging the Three Reads routine. This routine supports reading comprehension of problems and meta-awareness of mathematical language.



Did you start from 0 or 27.5 on the number line? How can you tell from the provided context?

10.H2.2 Collaboration: Real-World Absolute Value Inequalities (continued)

- d. Complete the table to determine if the given x -value is within the range of 4.2 minutes from the average.

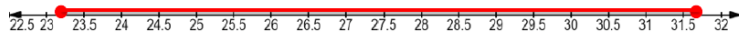
x	$ x - 27.5 $	Within 4.2 minutes?
22.9	4.6	No
23.4	4.1	Yes
30.2	2.7	Yes
31.7	4.2	Yes



What is the significance of a “Yes” vs. a “No” response in the last column? What does it mean in the context of this problem?

- e. Solve the inequality $|x - 27.5| \leq 4.2$.
 $23.3 \leq x \leq 31.7$

- f. Graph the solution on the number line.



- g. Compare the graph of the solution set of $|x - 27.5| \leq 4.2$ to the graph in Part C.
They are the same.

10.H2.3 Guided Instruction: Real-World Absolute Value Inequalities

Component Overview: Students will receive formal instruction on solving absolute value inequalities using two problems that are provided in a real-world context.



10.H2.3 Guided Instruction: Real-World Absolute Value Inequalities

1. Gretchen made a personal goal of earning \$100 per week at her after-school job. Each week, she has been averaging within \$14.25 of her goal.
 - a. Write an absolute value inequality that represents x , the amount of her weekly paycheck.
 $|x - 100| \leq 14.25$
 - b. Solve the inequality and write the answer using set-builder notation.
 $\{x | 85.75 \leq x \leq 114.25\}$

2. The length of a machine part should be 9.4 centimeters (cm). The manufacturing tolerance is within 0.05 cm.
 - a. Write an absolute value inequality that represents L , the length of the machine part.
 $|L - 9.4| \leq 0.05$
 - b. Solve the inequality and write the answer using set-builder notation.
 $\{L | 9.35 \leq L \leq 9.45\}$



For this activity, consider guiding students through question 1a as the absolute value inequality component, and then question 1b as the solution, using set-builder notation part. For question 2, provide the narrative information and release students to try question 2a on their own. Bring them back together to ensure they understand that component. Then, repeat this process for question 2b with the solution component.

10.H2.4 Guided Practice: Real-World Absolute Value Inequalities

Component Overview: Students will engage in two guided practice problems involving real-world absolute value inequalities.



10.H2.4 Guided Practice: Real-World Absolute Value Inequalities

1. Caleb is on the high school swim team. During the swim season, he had an average time of 96.1 seconds in the 100-meter freestyle. All his times were within 4.7 seconds of his average.

- a. Write an absolute value inequality that represents t , the time in seconds, it takes Caleb to swim the 100-meter freestyle.

$$|t - 96.1| \leq 4.7$$

- b. Solve the inequality.

$$\{t | 91.4 \leq t \leq 100.8\}$$

2. Tocara just leveled up on her video game. In order to achieve this accomplishment, she had to spend an average of 50 minutes each day playing. She spent less time playing on school days, but more time on the weekends. Each of her daily times was within 21 minutes of her average.

- a. Define the variable.

The number of minutes playing a video game on a given day.

- b. Write an absolute value inequality that represents the context.

$$|x - 50| \leq 21$$

- c. Solve the inequality. Write the solution as a compound inequality.

$$29 \leq x \leq 71$$

- d. Graph the solution on the number line.



For this activity, consider leveraging a Think-Pair-Share routine for question 1. Allow students a few minutes to try question 1 on their own. Then, have the students pair up with their partner to compare answers and come to a consensus on the answer. Bring the class back together to review question 1 to ensure any misconceptions are addressed, and release students to work on question 2 with or without their partner.

10.H2.5 Your Turn!

Component Overview: Students will complete a real-world absolute value inequality problem on their own.



10.H2.5 Your Turn!

Liana loves to play basketball. She plays on her high school team and averages 8.9 points per game. Her highest and lowest scoring games were within 1.4 points of her average. What is the range of Liana's points per game?

1. Define the variable.
The amount of points per game.
2. Write an absolute value inequality that represents the context.
 $|x - 8.9| \leq 1.4$
3. Solve the inequality and write the answer using set-builder notation.
 $\{x | 7.5 \leq x \leq 10.3\}$
4. Graph the solution on the number line.



Consider using this time to work with small groups of students who struggled with question 1b and 2 from 10.H2.4 Guided Practice. Students may need additional support with determining the necessary information from the context and writing the absolute value inequality correctly.

10.H2.6 Wrap-Up: Reflection

Component Overview: Students will complete a reflection on the learning targets from this lesson.



10.H2.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate how you feel about each learning target.

Answers will vary.

1. I can write and solve absolute value inequalities for a real-world context.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



2. I can represent solutions to absolute value inequalities algebraically or graphically.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



Prompt students to compare their Wrap-Up reflections from the unit. How have you grown this unit? Have you applied your growth mindset in this unit?